Multiple Linear Regression (MLR) - Mathematical Derivation

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Step 1: Set Up the Problem

Linear Model for One Observation:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_p x_{ip} + \epsilon_i$$

Where:

- y_i = observed target value for observation i
- β_0 = intercept
- $\beta_1, \beta_2, ..., \beta_p$ = coefficients for p features
- x_{ij} = j-th feature value for i-th observation
- ϵ_i = error term

Matrix Form (All Observations):

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Where:

- **y** (n×1): vector of observed values

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

- **X** (n×(p+1)): feature matrix with intercept column

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

- **β** (p+1×1): coefficient vector to estimate

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

- **ε** (n×1): error vector

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Step 2: Define Loss Function (RSS)

We want to **minimize the Residual Sum of Squares**:

Scalar Form:

$$RSS(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$$

Where:

- $\hat{y}_i = \mathbf{x}_i^T \boldsymbol{\beta}$ is the predicted value
- $\mathbf{x}_i$ is the i-th row of X (transposed to column vector)

Matrix Form:

$$RSS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

The **residuals** are:

$$\boldsymbol{\epsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$$

So:

$$RSS(\boldsymbol{\beta}) = \boldsymbol{\epsilon}^T\boldsymbol{\epsilon} = ||\boldsymbol{\epsilon}||^2$$

Step 3: Expand the Matrix Expression

Starting with:

$$RSS = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Expand using the formula **$(A - B)^T(A - B) = A^TA - A^TB - B^TA + B^TB$**:

$$RSS = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Simplification:

Since $\mathbf{y}^T\mathbf{X}\boldsymbol{\beta}$ and $\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y}$ are both **scalars** and equal:

$$\mathbf{y}^T\mathbf{X}\boldsymbol{\beta} = (\mathbf{y}^T\mathbf{X}\boldsymbol{\beta})^T = \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y}$$

Therefore:

$$RSS = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Step 4: Take Derivative with Respect to β

To minimize RSS, find where the gradient equals zero:

$$\frac{\partial RSS}{\partial \boldsymbol{\beta}} = \mathbf{0}$$

Applying Derivative Rules:

Rule 1: Derivative of constant = 0

$$\frac{\partial}{\partial \boldsymbol{\beta}}(\mathbf{y}^T\mathbf{y}) = \mathbf{0}$$

Rule 2: Derivative of linear term

$$\frac{\partial}{\partial \boldsymbol{\beta}}(-2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y}) = -2\mathbf{X}^T\mathbf{y}$$

Rule 3: Derivative of quadratic form

$$\frac{\partial}{\partial \boldsymbol{\beta}}(\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}) = 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Combining:

$$\frac{\partial RSS}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Step 5: Solve for β (Normal Equations)

Set the gradient equal to zero:

$$-2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$$

Divide both sides by 2:

$$-\mathbf{X}^T\mathbf{y} + \mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$$

Rearrange:

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$$

These are the NORMAL EQUATIONS

Step 6: Solve for β (Closed-Form Solution)

Starting from the normal equations:

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$$

Multiply both sides by $(\mathbf{X}^T\mathbf{X})^{-1}$ (assuming it exists):

$$(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Simplify:

Since $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{I}$ (identity matrix):

$$\mathbf{I}\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

FINAL SOLUTION (OLS Estimator):

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

What This Means:

- $\hat{\boldsymbol{\beta}}$ is the **Ordinary Least Squares (OLS)** estimator
- It minimizes the sum of squared residuals
- It gives us the **best linear unbiased estimator** under certain assumptions

Step 7: Verify It's a Minimum (Second Derivative Test)

Calculate the second derivative (Hessian matrix):

$$\frac{\partial^2 RSS}{\partial \boldsymbol{\beta}^2} = 2\mathbf{X}^T\mathbf{X}$$

Positive Definiteness:

For any non-zero vector **v**:

$$\mathbf{v}^T(\mathbf{X}^T\mathbf{X})\mathbf{v} = (\mathbf{X}\mathbf{v})^T(\mathbf{X}\mathbf{v}) = ||\mathbf{X}\mathbf{v}||^2 \geq 0$$

This is **positive semi-definite**, so our critical point is indeed a **minimum** ✓

Step 8: Make Predictions

Once we have the estimated coefficients $\hat{\boldsymbol{\beta}}$:

For All Training Data:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

This can be written as:

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$$

Where **H** is the **Hat Matrix**:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

For New Data Point x_0:

If we have a new observation with features $\mathbf{x}_0 = [1, x_{01}, x_{02}, ..., x_{0p}]^T$:

$$\hat{y}_0 = \mathbf{x}_0^T\hat{\boldsymbol{\beta}}$$

Quick Reference: All Key Equations

Component	Equation
Model	$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$
Loss (RSS)	$RSS = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$

Component	Equation
Expanded RSS	$RSS = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$
Gradient	$\frac{\partial RSS}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$
Normal Equations	$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$
OLS Solution	$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
Predictions	$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$
Residuals	$\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{y}}$

Notation Summary

Symbol	Meaning	Dimension
y_i	Target value for observation i	scalar
y	Vector of all target values	n × 1
x_{ij}	j-th feature of i-th observation	scalar
X	Feature matrix (including intercept)	n × (p+1)
β_j	Coefficient for j-th feature	scalar
β	Vector of all coefficients	(p+1) × 1
ϵ_i	Error for i-th observation	scalar
ε	Vector of all errors	n × 1
$\hat{y}_i$	Predicted value for observation i	scalar
$\hat{\mathbf{y}}$	Vector of all predictions	n × 1
$\hat{\boldsymbol{\beta}}$	Estimated coefficient vector	(p+1) × 1
H	Hat matrix	n × n
n	Number of observations	integer

Symbol	Meaning	Dimension
p	Number of features (excluding intercept)	integer

Key Assumptions for Valid Solution

1. **Invertibility:** $\mathbf{X}^T\mathbf{X}$ must be invertible
   - Implies **no perfect multicollinearity** among features
   - Requires $n > p$ (more observations than features)
2. **Linearity:** Relationship between X and y is linear
3. **Independence:** Observations are independent
   - $E[\epsilon_i \epsilon_j] = 0$ for $i \neq j$
4. **Homoscedasticity:** Constant error variance
   - $\mathrm{Var}(\epsilon_i) = \sigma^2$ for all i
5. **Normality:** Errors are normally distributed
   - $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$

Additional Performance Metrics

Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^{n} \hat{\epsilon}_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Total Sum of Squares (TSS):

$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Where $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is the mean of y.

R-squared (Coefficient of Determination):

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{TSS - RSS}{TSS}$$

- Interpretation: Proportion of variance explained by the model
- Range: [0, 1] (higher is better)

Adjusted R-squared:

$$\bar{R}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

- Penalizes adding unnecessary features
- More reliable when comparing models with different numbers of features

Mean Squared Error (MSE):

$$MSE = \frac{1}{n} RSS = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

Estimate of Error Variance:

$$\hat{\sigma}^2 = \frac{RSS}{n - p - 1}$$

The denominator (n-p-1) is the **degrees of freedom**:

- Subtract 1 for the intercept
- Subtract p for the p features

Example Calculation

Given Data:

```
3 observations, 2 features (plus intercept)

y = [10, 20, 30]ᵀ

X = [1  2   3]
    [1  4   5]
    [1  6   7]
```

Step 1: Calculate X^TX

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 3 & 12 & 15 \\ 12 & 56 & 72 \\ 15 & 72 & 83 \end{bmatrix}$$

Step 2: Calculate X^TY

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 60 \\ 280 \\ 340 \end{bmatrix}$$

Step 3: Solve Normal Equations

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

(Computed using matrix inversion)

Step 4: Make Predictions

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

Summary

Multiple Linear Regression finds the best-fitting hyperplane through n-dimensional data by:

1. **Setting up** a linear model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$
2. **Defining** a loss function: $RSS = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
3. **Taking derivatives** and setting to zero: $\frac{\partial RSS}{\partial \boldsymbol{\beta}} = 0$
4. **Solving** the normal equations: $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$
5. **Obtaining** the closed-form solution: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
6. **Making predictions** with: $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$

The solution is elegant, computationally efficient, and forms the foundation for understanding more advanced regression techniques.